

§ 5 全微分方程

👉 全微分方程

👉 积分因子

复习

若 $P(x, y), Q(x, y)$ 在单连通区域 G 内具有连续的一阶偏导数, 则 $\int_L Pdx + Qdy$ 与路径经无关

$$\Leftrightarrow \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \Leftrightarrow \oint_L Pdx + Qdy = 0 \quad \forall L \in G$$

$\Leftrightarrow Pdx + Qdy$ 是某个二元函数 $u(x, y)$ 的全微分
即, $\exists u(x, y)$ 使 $du(x, y) = Pdx + Qdy$

$$\text{且 } u(x, y) = \int_{(x_0, y_0)}^{(x, y)} Pdx + Qdy = \int_{x_0}^x P(x, y_0)dx + \int_{y_0}^y Q(x, y)dy$$

$$\text{或 } = \int_{y_0}^y Q(x_0, y)dy + \int_{x_0}^x P(x, y)dx - (*)$$

一、全微分方程

1. 定义

若一阶方程 $P(x, y)dx + Q(x, y)dy = 0$ --(1)

的左端恰好是某个二元函数 $u = u(x, y)$ 的全微分

即, $du = P(x, y)dx + Q(x, y)dy$

则称方程(1)为全微分方程或恰当方程.

2. 判定方法

显然: 方程(1)成为全微分方程 $\Leftrightarrow \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}, (x, y) \in G$

(G 单连通区域, P 、 Q 在 G 内有连续的一阶偏导数)

3.解法

若方程(1)是全微分方程 $\Rightarrow$ 方程(1)可写成
 $du(x, y) = 0$, 故(1)的通解为 $u(x, y) = C$ ——隐函数!

由以上讨论, 求方程 (1)的解归结为求 $u(x, y)$

方法1 应用曲线积分与路径无关.

由上面的复习 (*)式即可求出

$$\begin{aligned} u(x, y) &= \int_{(x_0, y_0)}^{(x, y)} Pdx + Qdy \\ &= \int_{x_0}^x P(x, y_0)dx + \int_{y_0}^y Q(x, y)dy \\ &\text{或} \\ &= \int_{y_0}^y Q(x_0, y)dy + \int_{x_0}^x P(x, y)dx \end{aligned}$$

方法2 利用偏积分法求 $u(x, y)$

$$\because du(x, y) = P(x, y)dx + Q(x, y)dy$$

$$\frac{\partial u}{\partial x} = P(x, y) \quad (I) \quad \frac{\partial u}{\partial y} = Q(x, y) \quad (II)$$

将(I)两边对 x 积分得

$$u(x, y) = \int P(x, y)dx + \varphi(y)$$

($\varphi(y)$ 任意函数) 将上式对 y 求偏导:

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y} \left[\int P(x, y)dx \right] + \varphi'(y) \stackrel{\text{由II}}{=} Q(x, y)$$

把 $\varphi(y)$ 确定出来, 就得方程 (1) 的解
用直接凑全微分的方法.

方法3

例1 解方程 $(x^2 + y)dx + (x + y)dy = 0$

解法1 $\because \frac{\partial P}{\partial y} = 1 = \frac{\partial Q}{\partial x}$

$\therefore$ 原方程为全微分方程

$$u(x, y) = \int_{(0,0)}^{(x,y)} (x^2 + y)dx + (x + y)dy$$

$$= \int_0^x x^2 dx + \int_0^y (x + y)dy = \frac{x^3}{3} + xy + \frac{y^2}{2}$$

$$\therefore \frac{x^3}{3} + xy + \frac{y^2}{2} = C$$

解法2 $\because \frac{\partial u}{\partial x} = x^2 + y \quad (1) \quad \frac{\partial u}{\partial y} = x + y \quad (2)$

对 (1) 积分得: $u(x, y) = \frac{x^3}{3} + xy + \varphi(y) \quad (3)$

将(3)对求 y 偏导:

$$\frac{\partial u}{\partial y} = x + \varphi'(y) \stackrel{\text{由(2)}}{=} x + y \Rightarrow \varphi'(y) = y$$

解得: $\varphi(y) = \frac{y^2}{2}$ (任取一原函数)

$$\therefore u(x, y) = \frac{x^3}{3} + xy + \frac{y^2}{2}$$

故通解为 $\frac{x^3}{3} + xy + \frac{y^2}{2} = C$

解法3 $(x^2 dx + y dy) + y dx + x dy = 0$

$$d \frac{x^3}{3} + d \frac{y^2}{2} + dxy = 0$$

即 $\frac{x^3}{3} + xy + \frac{y^2}{2} = C$ —— "分项组合"

凑全微分记住 :

$$xdy + ydx = d(xy) \quad xdx + ydy = \frac{1}{2}d(x^2 + y^2)$$

$$\frac{ydx - xdy}{y^2} = d\left(\frac{x}{y}\right) \quad \frac{ydx - xdy}{x^2 + y^2} = d\left(\arctan \frac{x}{y}\right)$$

$$\frac{xdy - ydx}{x^2} = d\left(\frac{y}{x}\right) \quad \frac{ydx - xdy}{yx} = d\left(\ln \frac{x}{y}\right)$$

$$\frac{ydx + xdy}{yx} = d(\ln xy)$$

等等.

二、积分因子

对 $P(x, y)dx + Q(x, y)dy = 0$ ——(1)

若 $\frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}$, 即方程 (1) 不是全微分方程

例如 $ydx - xdy = 0$ $\frac{\partial P}{\partial y} = 1 \neq \frac{\partial Q}{\partial x} = -1$

$\therefore$ 不是全微分方程

但 $\frac{1}{y^2}(ydx - xdy) = 0$, 即 $\frac{ydx - xdy}{y^2} = 0$

——是全微分方程

即 $d\left(\frac{x}{y}\right) = 0 \therefore \frac{x}{y} = c$ $\frac{1}{y^2}$ ——称为积分因子

定义 当 $\frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}$ 时, 若 $\exists$ 一个函数 $\mu(x, y)$

使 $\mu(x, y)P(x, y)dx + \mu(x, y)Q(x, y)dy = 0$

成为全微分, 则称 $\mu(x, y)$ 为方程(1)的积分因子.

注

积分因子一般不容易求得, 但在简单情形下可观察出来

例2 解方程 $ydx - xdy = 0$

解 $\frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}$

$$\mu(x, y) = \frac{1}{y^2} \quad d\left(\frac{x}{y}\right) = \frac{ydx - xdy}{y^2} \Rightarrow \frac{x}{y} = C$$

$$\mu(x, y) = \frac{1}{x^2} \quad d\left(\frac{-y}{x}\right) = \frac{ydx - xdy}{x^2} \Rightarrow -\frac{x}{y} = C$$

$$\mu(x, y) = \frac{1}{xy} \quad d\left(\ln \frac{x}{y}\right) = \frac{ydx - xdy}{xy} \Rightarrow \ln \frac{x}{y} = C$$

例3 解方程 $(y + 2xy^2)dx + (x - 2x^2y)dy = 0$

解 $\frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}$

$$\text{分项: } (ydx + xdy) + (2xy^2dx - 2x^2ydy) = 0$$

$$d(xy) + (y^2dx^2 - x^2dy^2) = 0$$

$$\mu(x, y) = \frac{1}{x^2y^2} \quad \frac{1}{(xy)^2} d(xy) + \left(\frac{1}{x^2} dx^2 - \frac{1}{y^2} dy^2 \right) = 0$$

$$d\left(-\frac{1}{xy}\right) + d(\ln x^2 - \ln y^2) = 0$$

$$\therefore -\frac{1}{xy} + \ln \frac{x^2}{y^2} = C$$